

Digit Recognizer

CNN trained on MNIST with a live drawing canvas

★ MASTERS STANDOUT

Masters

Portfolio

Proud

Repo / Path	asadbek/AI ML/2-project
Domain	Computer vision / deep learning
Stack	PyTorch CNN + torchvision (MNIST), FastAPI; React 18 + Vite + Tailwind
Status	Working - ~90% complete; ~99% test accuracy
Live / Deploy	Not publicly deployed

Overview

A handwritten-digit recogniser: draw on a browser canvas and a CNN predicts the digit with live confidence bars. The model is a real convolutional network trained on the actual MNIST dataset.

Key Features & Technical Highlights

- 2 conv blocks with BatchNorm + Dropout and a 512-unit dense head.
- Trained on the real 60k/10k MNIST split with affine data augmentation and best-checkpoint saving.
- Documented preprocessing contract (center-of-mass 28x28) shared across train and serve.
- Clean train/serve split; trained weights (mnist_model.pth) committed.

Engineering Evidence & Metrics

- Real model training on a real dataset with augmentation; ~99% reported test accuracy.
- Per-class accuracy reporting.

Why It Matters (application / interview talking points)

- Strong, classic ML evidence: you trained a model (not just called an API) and reported rigorous metrics.